

Assignment

CSE 251

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Assignment

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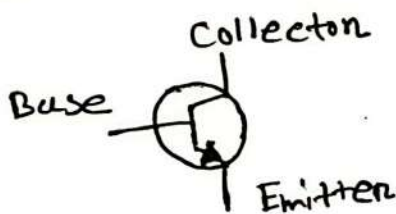
Answer to the question No-01

A bipolar Junction transistor consists of two P-n Junction formed by sandwiching either P-type or n-type semiconductor between a pair of opposite types, so there are two types of bipolar junction transistor (BJT)

1. n-p-n transistor

2. P-n-p transistor

1. n-p-n transistor



n	p	n
e e e e	o o	e e e
e e e e	o o	e e e
e e e e	o o	e e e
e e e e	o o	e e e
e e e e	o o	e e e
e e e e	o o	e e e

Thickness:

Collector > Emitter > Base

Doping:

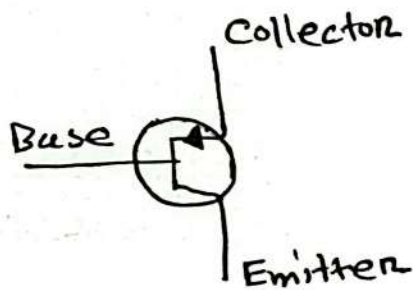
Emitter > collector > Base

Majority carriers:

Electrons

Minority carriers: Holes

2. P-n-p transistor



P	n	P
o o o o	ee	o o o
o o o o	ee	o o o
o o o o	ee	o o o
o o o o	ee	o o o
o o o o	ee	o o o
o o o o	ee	

Thickness: $\text{Collector} > \text{Emitter} > \text{Base}$

Doping: $\text{Emitter} > \text{collector} > \text{Base}$

Silicon vs Germanium: No changing from silicon to germanium does not alter the symbols or basic structure. However the electrical characteristics change. Ge has a lower base-emitter turn on voltage of $\approx 0.3 \text{ V}$ compared to Si $\approx 0.7 \text{ V}$

Answer to the Question No - 2

Given, $I_E = 8 \text{ mA}$, $I_B = \frac{I_C}{100}$

We know,

$$I_E = I_B + I_C$$

$$\Rightarrow I_E = \frac{I_C}{100} + I_C$$

$$\Rightarrow I_E = \frac{I_C + 100I_C}{100}$$

$$\Rightarrow 100 I_E = 101 I_C$$

$$\begin{aligned}\Rightarrow I_C &= \frac{100 I_E}{101} \\ &= \frac{100 \times 8}{101} \\ &= 7.92 \text{ mA}\end{aligned}$$

$$\therefore I_C = 7.92 \text{ mA}$$

$$\begin{aligned}\text{Again, } I_B &= \frac{I_C}{100} = \frac{7.92}{100} \\ &= 0.0792 \text{ mA}\end{aligned}$$

$$\therefore I_B = 0.0792 \text{ mA}$$

Answer to the Question No: 3

a) From the graph:

$$\text{For, } V_E = 0.7V$$

$$I_E = 0mA$$

$$\text{and, } V_E = 0.8V$$

$$I_E = 5mA$$

$$\begin{aligned}\therefore \text{Average ac resistance } r_e &= \frac{\Delta V_{BE}}{\Delta I_E} \\ &= \frac{(0.8 - 0.7)}{(5 - 0) \times 10^{-3}} \\ &= 20\Omega\end{aligned}$$

b) yes, the approximation is valid. Since 20Ω is negligible compared to resistances in the kilohm range, ignoring it (treating it as ∞ as shown in Fig: 3.10c) will not introduce significant error in the circuit calculations.

Answer to the Question No-4

Define I_{CBO} (Collector - Base open):

The leakage current flowing from the collector to the base with the emitter terminal left open. It is the reverse saturation current of the collector - base junction. ($I_E = 0$)

Define I_{CEO} (Collector - Emitter open):

The leakage current flowing from the collector to the emitter with the base terminal left open. ($I_B = 0$).

Relationship: $I_{CEO} = (\beta + 1) I_{CBO}$ [β = current gain]

Difference: I_{CBO} is the collector - base leakage current measured with the emitter open, whereas I_{CEO} is the collector - emitter leakage current measured with the base open and is much larger because it is amplified by the transistor's current gain (β).

Magnitude: I_{CEO} is significantly larger than I_{CBO} because the transistor amplifies I_{CBO} by the current gain (β) which is typically (50-200).

Answer to the question No-5

a) Given, $\alpha_{dc} = 0.98$

we know,

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}} = \frac{0.98}{1 - 0.98} = 49$$

b) Given, $\beta_{dc} = 120$

we know,

$$\alpha_{dc} = \frac{\beta_{dc}}{\beta_{dc} + 1} = \frac{120}{120 + 1} = 0.992$$

c) Given,

$$\beta_{dc} = 120, \quad I_e = 2 \text{ mA}$$

we know,

$$I_B = \frac{I_e}{\beta_{dc}} = \frac{2 \times 10^{-3}}{120} = 0.0167 \text{ mA}$$

$$\text{and, } I_E = I_e + I_B = 2 + 0.0167$$

$$= 2.0167 \text{ mA.}$$